

GPU Architecture

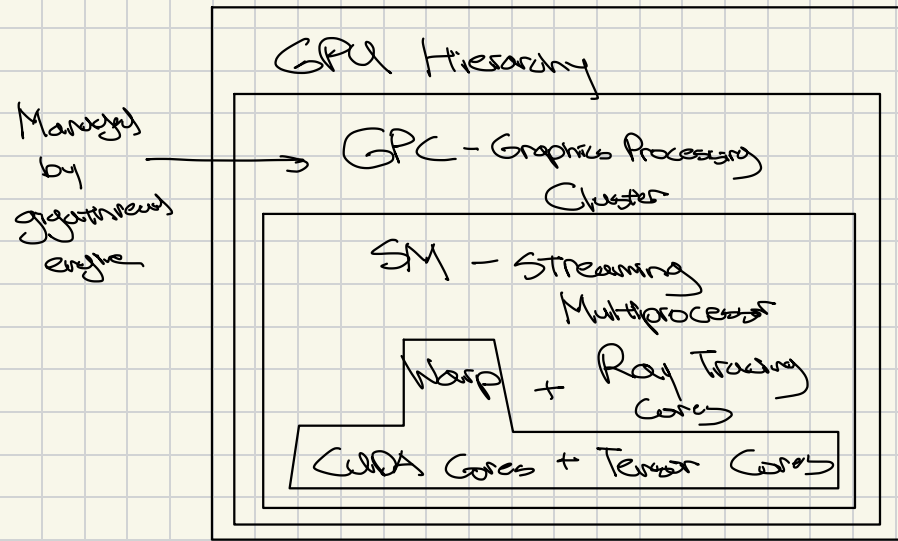
GPU

vs

CPU

- more cores w/ slower execution rate
- limited to certain tasks

- less cores w/ faster execution rate
- flexible, can run OS etc



Cores execute all calculations of the GPU

Cuda Cores - ALUs

- Mostly fused multiply & add ($A \times B + C$)
- Either 32 bit FP or INT
- One multiply + add per clock cycle
- A few special functions units per SM
 - Div / sqrt / trig

Tensor Cores

- Does $A \times B + C = D$ concurrently
 - Parallel mult
 - Addition with parallel reduction tree

Memory Architecture

- Data constantly streamed from SDRAM to L2 cache on chip
 - Done by memory controllers
- Newest gen can encode in multiple voltages
 - Binary to Ternary - ~~like~~ more info
 - PAM 3/4

SIMD

- Single Instruction Multiple Data
- Same instruction on a lot of numbers
 - Ex. Transforming and moving object
 - Same instruction to move origin also used to move rest of object
- Like one opcode used on multiple lanes with diff data

Thread Architecture

- 1 instruction per thread
- bundle of 32 threads $\rightarrow$ warp $\rightarrow$ all have same instruction (SIMD)
- warps grouped into thread blocks handled by SM
- Thread blocks grouped into grids

SIMT (newer architecture)

- Single instruction multiple threads
- Individual threads can progress at diff rates
- L1 cache shared by all threads in SM
- More flexibility dealing with warp divergence
 - If branching based on data
 - Still able to progress and synchronize instead of stalling on one warp